

# llmXive Automated Review of Geometric Action Model for Robot Policy Learning

*llmXive automated review panel*

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**Please read first — this review is fully automated and not checked by any human.** It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

## How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

**This paper's panel (9 lenses):** Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

## Claim Accuracy

*Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and their support by the provided evidence

and citations.

**1. Inference Speed Claim (55x Faster):** The paper repeatedly claims GAM is “55x faster” than baselines (Abstract, Introduction, Section 4.2). The Appendix (Table D.1) reveals a methodological inconsistency in this comparison. GAM’s 6.9ms latency is achieved using **CUDA Graphs**, whereas the baseline latencies (e.g.,  $\pi_{0.5}$  at 29.2ms, Cosmos Policy at 382.4ms) are reported **without** CUDA Graphs (indicated by the  $\times$  in the “CUDA Graphs” column). While the authors note in the appendix that GAM’s deployment setting uses this optimization, the main text presents the 55x figure as a general architectural advantage over “diffusion-based Cosmos Policy” without qualifying that the baseline was not optimized with the same deployment technique. A fair comparison would either apply CUDA Graphs to the baselines (if supported) or explicitly state that the speedup is relative to the specific unoptimized baseline configurations. The current phrasing overstates the architectural speed advantage by conflating it with deployment-specific optimizations.

**2. Camera Perturbation Performance Claim:** The Introduction claims GAM achieves “outstanding performance in camera perturbation settings ( $\uparrow 9.7\%$ ).” Table 1 (LIBERO-Plus) shows GAM at 83.1% success rate in the “Cam.” column. The closest WAM baseline, Cosmos-Policy, is at 73.4%, yielding a 9.7% point gap. However, the VLA baseline  $\pi_{0.5}$  is at 72.0%. The claim is technically accurate regarding the gap to the *best* baseline (Cosmos), but the phrasing “outstanding performance... which requires geometric understanding priors” implies a universal superiority over all foundation models. The gap to the best VLA ( $\pi_{0.5}$ ) is 11.1%, which is larger, but the text does not distinguish between the types of baselines in this specific claim. While not a factual error, the claim would be more precise if it specified “outperforming the best video-based world model baseline by 9.7%.”

**3. Dataset Scale Claim:** The Introduction and Section 4.1 state the model is pretrained on “784K single-arm robot trajectories.” The Appendix (Section A.1) details the data sources (OXE, MimicGen, RoboCasa365) and their ratios (72%, 18%, 10%) but does not explicitly confirm the total count of 784K. While this number is likely the result of the authors’ filtering process, the claim is a specific factual assertion that should be explicitly tied to the filtered dataset count in the text or appendix to ensure the reader can verify the scale of the training data.

**4. Citation Support:** The citations for baseline methods (e.g.,  $\pi_{0.5}$ , Cosmos-Policy, Open-VLA) are generally accurate and correspond to the correct arXiv preprints or papers listed in the bibliography. The claim that “3D cues... are left implicit” in VLAs is supported by the cited literature (e.g., \cite{fei2025libero} which analyzes VLA robustness). The claim that GFMs are used as “static feature extractors” in prior work is supported by the cited papers (e.g., \cite{li2025spatial}, \cite{sun2026rocket}). No unsupported citations were found.

**Conclusion:** The paper’s core scientific claims are supported by the data, but the specific quantitative claims regarding inference speed and the precise nature of the performance gap in camera perturbations require minor clarification to avoid overstatement or ambiguity regarding the comparison conditions.

#### Action items.

- **[writing]** The ‘55x faster’ claim (Abstract, Intro) compares GAM with CUDA Graphs (6.9ms) against baselines without it. Clarify that this speedup includes deployment optimizations not

applied to baselines, or re-evaluate baselines with the same settings for a fair architectural comparison.

- **[writing]** The ‘9.7%p’ camera perturbation gain (Intro) compares GAM to Cosmos-Policy (73.4%) but ignores  $\pi_{0.5}$  (72.0%). Specify that the gain is relative to the best WAM baseline or clarify the comparison scope to avoid implying a universal lead over all foundation models.
- **[writing]** The ‘784K trajectories’ claim (Sec 4.1) lacks explicit confirmation in the appendix. State the exact filtered dataset count or link the 784K figure to the specific OXE/Mimic-Gen/RoboCasa365 split to ensure reproducibility of the data scale.

## Figure Critic

*Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_figure\_critic.md

### Figure 1

The figure presents a dataset mixture but contains a critical contradiction: the bar chart values are labeled as percentages in the caption but numerically represent raw counts (scaled by 10,000), which conflicts with the pie chart’s percentage labels and the stated axis definition.

### Figure 2

Figure 2 visually demonstrates a robot task progression but lacks sufficient caption detail to explain the multi-view layout and temporal structure, potentially hindering interpretation of the prediction results.

### Figure 3

Figure 3 effectively presents zero-shot robustness results across eight perturbation categories and an average summary. The subplots are clearly labeled, axes are consistent, and the legend is present and readable in each panel, supporting the caption’s claims.

### Figure 4

Figure 4 effectively illustrates the GAM pipeline and quantitative advantages, but contains several spelling errors (‘Geometric’, ‘Pertrubed’) and uses a reversed x-axis for latency that may confuse readers.

## Figure 5

Figure 5 provides a clear and detailed schematic of the Geometric Action Model (GAM) architecture, effectively illustrating the flow from observation encoding through the causal transformer to action decoding. The diagram is well-organized with distinct color-coded sections and includes a helpful key for the attention mask visualization in panel (b).

## Figure 6

Figure 6 effectively illustrates the architectural differences between Video WAMs, Geometry-aware VLAs, and the proposed GAM. The diagram is clear, well-labeled, and the caption accurately describes the three distinct approaches shown.

## Figure 7

The figure effectively illustrates the experimental setup and camera views, but the sub-figure labels (a) and (b) and the terms ‘In-distribution’ and ‘Out-of-distribution’ lack explicit definitions in the caption.

## Figure 8

Figure 8 effectively illustrates four distinct real-world manipulation tasks using a grid of sequential photographs. The visual content aligns perfectly with the caption, and the embedded text labels clearly describe the specific objective for each task.

## Figure 9

Figure 9 presents attention visualizations but fails to provide a legend or colorbar to interpret the heatmap values. Additionally, the visual content (scene heatmaps) appears to contradict the caption’s claim of visualizing ‘action tokens’ specifically, creating ambiguity about what is being shown.

## Figure 10

The figure presents a clear visual layout of robot tasks and results, but the caption’s definition of the bar colors (in-domain vs. out-of-domain) directly contradicts the figure’s internal legend (Ours vs. Baselines), making the data interpretation impossible without guessing.

**Action items.**

- **[science]** Figure 1: The bar chart values (e.g., 18.0 for MimicGen) contradict the pie chart legend (18% for MimicGen) and the caption’s claim that the bar chart shows ‘percentage’. The sum of bar values is  $\sim 100$ , implying the bars show raw counts (in 10Ks) rather than percentages, creating a direct conflict with the caption and the pie chart’s percentage labels.
- **[writing]** Figure 2: The caption ‘LIBERO-Spatial: bowl from table center to plate’ is too brief and does not explain the four rows (Current RGB, Current Depth, GT Future, Pred Future) or the time steps ( $T=0$  to  $T=10$ ) shown in the figure.

- **[science]** Figure 2: The ‘Current RGB’ row shows a top-down view of the scene, while the ‘Current Depth’ and future rows show a side-view depth map; this perspective shift is not explained in the caption and may confuse readers about the spatial relationship between views.
- **[writing]** Figure 4: The x-axis of the ‘Inference Latency’ scatter plot is reversed (500 to 5), which is non-standard and potentially confusing.
- **[writing]** Figure 4: The x-axis label for the bar chart is misspelled as ‘Pertrubed Camera’ instead of ‘Perturbed Camera’.
- **[writing]** Figure 4: The title ‘Geometric Foundation Model’ is misspelled as ‘Geometric Foundation Model’ (missing ‘r’ in Geometric) in the top left.
- **[writing]** Figure 7: The sub-figure labels (a) and (b) in the right column are not defined in the caption, which only describes the general setup and ID vs. OOD camera setup without mapping these labels to specific camera views.
- **[writing]** Figure 7: The terms ‘In-distribution’ and ‘Out-of-distribution’ are used as labels for the robot end-effector views on the left, but the caption does not explicitly define what constitutes the in-distribution versus out-of-distribution setup.
- **[science]** Figure 9: The caption describes ‘Attention visualizations of action tokens,’ but the figure displays a grid of robot manipulation frames with heatmaps overlaid on the scene. It is unclear if these heatmaps represent attention over action tokens or spatial attention over the image; the visualization does not match the description of token-level attention.
- **[writing]** Figure 9: The figure lacks a legend or colorbar to interpret the heatmap intensity (e.g., attention weight or probability), making the visualizations unquantifiable.
- **[writing]** Figure 9: The row labels (‘Layer 13’, ‘Layer 26’, etc.) and column labels (‘t=1’, ‘t=3’, etc.) are present, but there is no explicit legend defining what the specific colors in the heatmaps represent.
- **[science]** Figure 10: The caption states that light bars represent in-domain and dark bars represent out-of-domain settings, but the bar chart legend identifies the colors as ‘Ours’, ‘ $\pi_{0.5}$ ’, and ‘Spatial Forcing’. This creates a direct contradiction where the visual encoding (color) conflicts with the textual description of the experimental conditions.
- **[science]** Figure 10: The ‘Stack milk and cube’ task shows a 100% success rate for ‘Ours’ (80+20), yet the ‘Pick and Place’ task (a prerequisite skill) shows a lower success rate for the same method. The stacked bars imply a breakdown of success types (e.g., ID vs OOD) but the caption’s definition of the bars does not align with the stacked nature of the data visualization.

## Jargon Police

*Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_jargon\_police.md

The manuscript relies heavily on domain-specific acronyms and mathematical shorthand that hinder accessibility for a broader robotics or computer science audience.

First, the Abstract introduces three major acronyms—**VLA** (vision-language-action models), **WAM** (video world-action models), and **GAM** (Geometric Action Model)—without defining them. While **GAM** is defined, **VLA** and **WAM** are used immediately. The Introduction defines **VLA** and **WAM** but repeats the acronym **GAM** without re-defining it in the context of the Abstract’s summary. The term **GFM** (Geometric Foundation Model) is also introduced in the Abstract without definition. Standard practice requires defining these at their very first occurrence in the Abstract or the first paragraph of the Introduction.

Second, the phrase “(↑9.7%p)” in the Introduction and Section 4.1 uses the symbol ‘p’ to denote percentage points. This is non-standard and potentially confusing; “percentage points” should be written out or the symbol ‘pp’ used if space is strictly limited, though ‘p’ is generally discouraged in formal writing.

Third, technical architectural terms are used without definition. In Section 3, the **DPT** head is mentioned without expansion (Dense Prediction Transformer). In Section 4.4, **key-value caching** is referenced as a standard inference technique without explanation. Additionally, the notation **SE(3)** in Section 3 is used without a brief textual explanation (e.g., “3D rigid body transformations”), which may alienate readers from control backgrounds who are less familiar with Lie group notation.

Finally, the term **block-causal self-attention** in Section 4.2 is a specific variant of attention mechanisms. While the paper cites a source, a brief parenthetical explanation (e.g., “attention that prevents future leakage within blocks”) would improve clarity for non-specialists. These changes are necessary to ensure the paper’s contributions are accessible beyond the immediate sub-field of geometric foundation models.

### Action items.

- **[writing]** The manuscript relies heavily on domain-specific acronyms and mathematical shorthand that hinder accessibility for a broader robotics or computer science audience. First, the Abstract introduces three major acronyms—**VLA** (vision-language-action models), **WAM** (video world-action models), and **GAM** (Geometric Action Model)—without defining them. While **GAM** is defined, **VLA** and **WAM** are used immediately. The Introduction defines **VLA** and **WAM** but repeats the acronym **GAM** without re-defining it in the context

## Logical Consistency

*Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_logical\_consistency.md

The logical flow of the paper is generally sound, with the core premise—that repurposing a Geometric Foundation Model (GFM) for temporal prediction improves 3D robustness—supported by the experimental design. The causal link between the architectural modification (inserting a predictor at the split layer) and the observed robustness gains in camera-perturbation settings is well-motivated by the hypothesis that 3D priors resolve spatial ambiguities that 2D models cannot.

However, there are two areas where the logical consistency between claims and evidence requires tightening:

1. **Speed Claims and Deployment Variables:** The introduction and analysis sections claim GAM is “55x faster” than diffusion-based baselines. This figure is derived from the deployment setting in Appendix Table 1 (6.9ms for GAM with CUDA Graphs vs. 382.4ms for Cosmos). The logical gap lies in attributing this massive speedup primarily to the *architecture* (single-pass vs. diffusion) while the comparison includes a significant *deployment* variable (CUDA Graphs enabled for GAM but not baselines). The appendix shows that without CUDA Graphs, GAM is 17.5ms, which is still faster than baselines but not 55x faster than the 382ms diffusion baseline (which is inherently slow). The claim “55x faster” conflates architectural efficiency with specific inference optimizations. The conclusion that the architecture is “substantially faster” is valid, but the specific magnitude of the claim is logically contingent on the deployment stack, not just the model design.
2. **Ablation Interpretation:** In Section 4.3 (Ablation Study), the authors state that removing the future-prediction losses ( $\mathcal{L}_{\text{depth}}$  or  $\mathcal{L}_{\text{feat}}$ ) has “minimal impact” on robustness when pretraining is used. Yet, they immediately argue that these losses are crucial for robustness when pretraining is *not* used. While this is a valid observation about the interaction between pretraining and auxiliary losses, the phrasing “minimal impact” followed by “substantially improve” creates a slight logical tension. It suggests the losses are redundant in the final setup but essential for the training process to work without pretraining. The paper should more explicitly frame this as: “The pretrained GFM encodes sufficient geometric dynamics such that explicit future prediction losses become redundant for robustness, whereas they are critical when the backbone lacks these priors.” This clarifies that the *necessity* of the mechanism is conditional on the initialization state, rather than the mechanism itself being universally non-impactful.

Overall, the causal mechanisms proposed (GFM priors  $\rightarrow$  better 3D reasoning  $\rightarrow$  robustness) are consistent with the data, but the quantitative claims regarding speed and the interpretation of ablation dependencies need to be more precise to avoid overgeneralization.



**Action items.**

- **[writing]** Speed Claims and Deployment Variables: The introduction and analysis sections claim GAM is “55x faster” than diffusion-based baselines. This figure is derived from the deployment setting in Appendix Table 1 (6.9ms for GAM with CUDA Graphs vs. 382.4ms for Cosmos). The logical gap lies in attributing this massive speedup primarily to the *architecture* (single-pass vs. diffusion) while the comparison includes a significant *deployment* variable (CUDA Graphs enabled for GAM but not baselines). The
- **[writing]** Ablation Interpretation: In Section 4.3 (Ablation Study), the authors state that removing the future-prediction losses ( $\mathcal{L}_{\text{depth}}$  or  $\mathcal{L}_{\text{feat}}$ ) has “minimal impact” on robustness when pretraining is used. Yet, they immediately argue that these losses are crucial for robustness when pretraining is *not* used. While this is a valid observation about the interaction between pretraining and auxiliary losses, the phrasing “minimal impact” followed by “substantiall

**Overreach**

*Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_overreach.md

The paper makes several strong claims regarding the superiority of the Geometric Action Model (GAM) over existing baselines, specifically that it is “more accurate, more robust, faster, and lighter.” While the experimental results support the claim of improved robustness (particularly in camera-perturbation settings), the other claims exhibit overreach when scrutinized against the provided data.

First, the claim of being “55x faster” (Introduction, Section 4.2) is derived from a comparison where GAM utilizes CUDA Graphs for inference optimization, while the baselines are evaluated using their “official PyTorch” paths without this specific optimization (Appendix, Section A.4). This is a methodological inconsistency that inflates the reported speedup. The 55x figure represents the gap between an optimized GAM and unoptimized baselines, not an inherent architectural advantage. The authors must either benchmark all methods with identical optimization stacks (including CUDA Graphs) or temper the claim to reflect the specific conditions under which this speedup was achieved.

Second, the assertion that GAM is “more accurate” (Abstract, Introduction) is not universally supported by the data. In Table 1 (LIBERO Original), GAM achieves a 97.6% success rate, which is lower than Cosmos-Policy (98.5%) and comparable to pi0.5 (96.9%). GAM’s accuracy advantage is primarily evident in the LIBERO-Plus (OOD) settings, particularly the camera perturbation. Claiming general “higher accuracy” without qualifying it as “higher accuracy in out-of-distribution scenarios” is an overstatement of the results.

Finally, the claim that GAM is “lighter” is imprecise. While GAM has fewer *trainable* parameters (~983M) compared to the full baselines, its total parameter count (1.4B) is not



significantly “lighter” than all competitors (e.g., Cosmos-Policy is 2B, but pi0.5 is 3.3B). The term “lighter” often implies total model size or memory footprint. A more accurate claim would be “more parameter-efficient” or “requires fewer trainable parameters,” which is supported by the ablation showing the backbone is largely frozen.

These issues do not invalidate the core contribution but require precise rephrasing to ensure the claims align strictly with the evidence presented.

#### Action items.

- **[science]** The claim of being ‘55x faster’ (Introduction, Sec 4.2) relies on a specific deployment configuration (CUDA Graphs) for GAM that is not applied to baselines. The text states baselines use ‘official PyTorch’ paths without CUDA Graphs, while GAM uses them. This comparison is unfair and overstates the speed advantage. The authors must either apply CUDA Graphs to all baselines or rephrase the claim to reflect the specific optimization used.
- **[writing]** The claim that GAM is ‘more accurate’ (Abstract, Introduction) is an overgeneralization. Table 1 shows GAM is not the most accurate on the standard LIBERO benchmark (Orig. SR 97.6% vs Cosmos-Policy 98.5% and pi0.5 96.9% is close, but Cosmos is higher). GAM is only ‘more accurate’ on specific perturbation settings (e.g., Camera). The text should qualify ‘more accurate’ to ‘more robust under perturbations’ or ‘more accurate in OOD settings’.
- **[writing]** The statement that GAM is ‘lighter’ (Abstract, Introduction) is ambiguous. While the total parameter count (1.4B) is lower than some baselines (e.g., 7B OpenVLA), it is comparable to or higher than others (e.g., 2B Cosmos, 3.3B pi0.5). The claim should be qualified to ‘fewer trainable parameters’ (which is true, ~983M vs 3.3B+) rather than ‘lighter’ in a general sense, as the frozen backbone still contributes to the model size.

## Safety Ethics

*Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_safety\_ethics.md

The paper presents a robot policy learning framework (GAM) that repurposes geometric foundation models for manipulation. From a safety and ethics perspective, the work is generally sound but requires specific clarifications regarding human data usage and the safety implications of high-speed control.

**Human Data and Consent:** In the Appendix (Section A.3, “Real-World Experiments Details”), the authors state: “We collected teleoperated demonstrations for each task: 284, 202, 184, and 169 demonstrations, respectively.” While the tasks appear to be standard manipulation (e.g., “Pick and place”), the collection of human demonstration data typically requires Institutional Review Board (IRB) approval and informed consent from the participants, especially if the data is to be published or used for training public models. The manuscript currently lacks a

statement confirming that ethical approval was obtained and that participants provided informed consent. A brief statement in the “Real-World Experiments Details” or “Acknowledgments” section addressing this is necessary to comply with standard research ethics guidelines.

**Data Privacy and Licensing:** The pre-training phase utilizes the Open-X Embodiment dataset (Appendix A.1). While this is a public dataset, the authors should explicitly confirm that their usage adheres to the dataset’s license terms and that no personally identifiable information (PII) or sensitive data was inadvertently included or exposed in their processed training set. A sentence confirming compliance with the original data licenses and privacy protocols would strengthen the ethical standing of the work.

**Safety Implications of High-Frequency Control:** The paper highlights a significant performance gain, achieving 145 Hz control (6.9 ms latency) and a 55x speedup over diffusion baselines (Section 4.2, “Inference Speed and Model Size”). While this efficiency is a major contribution, it introduces potential safety risks. High-frequency open-loop execution (predicting 8 steps and executing them without intermediate feedback, as noted in Section 4.1) in a physical environment can be hazardous if the model encounters an unexpected obstacle or state not present in the training distribution. The paper does not discuss the deployment of external safety layers (e.g., collision avoidance, emergency stop mechanisms, or safety filters) that would mitigate risks if the model’s geometric prediction fails. Given the “contact-rich manipulation” focus mentioned in the abstract, the authors should briefly address how safety is ensured during real-world deployment, particularly regarding the open-loop nature of the action chunks.

**Dual-Use Considerations:** The methodology is general-purpose and applicable to various robotic platforms. While the paper focuses on benign manipulation tasks (e.g., kitchen tasks, block stacking), the ability to learn robust policies from geometric priors could theoretically be adapted for more sensitive applications. However, the current scope and benchmarks do not suggest an immediate dual-use risk that would warrant a rejection, provided the standard safety mitigations mentioned above are acknowledged.

In summary, the paper is scientifically sound but requires minor revisions to explicitly address ethical clearance for human data collection, data privacy compliance, and the safety architecture surrounding high-speed open-loop control.

#### Action items.

- **[writing]** The real-world experiments section (Appendix A.3) describes collecting teleoperated demonstrations but lacks explicit confirmation of informed consent from human operators and IRB approval status. Given the use of human data for training, a statement on ethical clearance and consent procedures is required.
- **[writing]** The paper claims a 55x speedup and 145Hz control rate. While efficiency is positive, the authors must explicitly address safety implications of high-frequency open-loop execution (Section 4.1) in the absence of real-time safety monitors or collision avoidance layers, particularly for contact-rich tasks.
- **[writing]** The training data includes Open-X Embodiment (Appendix A.1). The authors should clarify if this dataset contains any personally identifiable information (PII) or sensitive data and confirm that their usage complies with the original dataset licenses and privacy

policies.

## Scientific Evidence

*Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_scientific\_evidence.md

The scientific evidence supporting the core claims of the Geometric Action Model (GAM) is generally strong, particularly regarding the architectural novelty and the simulation results on LIBERO-Plus. The ablation studies effectively isolate the contribution of the split layer  $L_s$  and the pretraining stage. However, there are specific concerns regarding the statistical rigor of the real-world evaluation and the fairness of the inference speed comparison that require clarification before the claims can be fully accepted.

First, the claim of a “55x” speedup over diffusion-based baselines (Introduction, Sec 4.2) is potentially misleading due to an asymmetry in the evaluation protocol. Table A.5 in the appendix reveals that the 6.9ms latency for GAM is achieved using CUDA Graphs, a deployment optimization not enabled for the baselines (which are measured at 29.2ms–382.4ms without CUDA Graphs). When comparing under matched conditions (Torch Compile only, no CUDA Graphs), the speedup reduces to approximately 1.7x (17.5ms vs 29.2ms for  $\pi_{0.5}$ ). While GAM is still faster, the “55x” figure conflates architectural efficiency with specific deployment optimizations. The main text should either present the matched comparison as the primary metric or explicitly qualify the 55x claim as a “best-case deployment scenario” to avoid overstating the inherent algorithmic advantage.

Second, the real-world robustness claims rely on a sample size that is statistically fragile. The authors report results from “20 trials per task” (10 nominal, 10 perturbed) in Sec 4.2 and Appendix A.3. In real-robot manipulation, success rates can exhibit high variance due to unmodeled dynamics, lighting changes, or minor hardware inconsistencies. With only 10 trials in the out-of-distribution (OOD) condition, a difference of a few successes (e.g., 7/10 vs 5/10) can lead to large percentage swings that may not be statistically significant. The paper currently lacks confidence intervals, standard errors, or a statistical significance test (e.g., a bootstrap test or Fisher’s exact test) to validate that the observed improvements over baselines like  $\pi_{0.5}$  are not due to random chance. Given the high stakes of “real-world” claims, a larger sample size (e.g., 50+ trials) or rigorous statistical reporting is necessary.

Finally, the interpretation of the ablation study in Table 3 requires nuance. The authors state that removing the future-prediction losses ( $\mathcal{L}_{\text{depth}}$ ,  $\mathcal{L}_{\text{feat}}$ ) has “minimal impact” on robustness when pretraining is present (89.7% vs 89.0% on LIBERO-Plus). However, they simultaneously claim these losses “substantially improve robustness” when pretraining is absent (73.4% vs 50.0%). This suggests the benefit of the geometric world modeling objective is conditional on the lack of a strong pretrained prior. The discussion should explicitly address this interaction: does the GFM backbone already encode sufficient geometric dynamics that the explicit prediction losses

become redundant in the pretraining regime? Clarifying this would strengthen the argument for when and why the proposed architecture is necessary.

#### Action items.

- **[science]** The claim of ‘55x faster’ inference (Sec 4.2) relies on a specific deployment configuration (CUDA Graphs) for GAM that is not applied to baselines. Table A.5 shows the gap narrows to  $\sim 1.7\times$  (17.5ms vs 29.2ms) without CUDA Graphs. The main text must clarify this asymmetry to avoid overstating the speed advantage.
- **[science]** Real-world robustness claims (Sec 4.2) are based on only 20 trials per task (10 ID, 10 OOD). This sample size is insufficient to robustly claim ‘substantial’ outperformance over baselines given the high variance typical in real-robot manipulation. Please report confidence intervals or conduct a statistical significance test (e.g., bootstrap) to support the conclusion.
- **[science]** The ablation study in Table 3 suggests removing `L_depth` and `L_feat` has ‘minimal impact’ on robustness (Plus SR 89.0% vs 89.7%). However, the text claims these losses ‘substantially improve robustness’ when pretraining is absent. The manuscript should explicitly reconcile these findings: does the benefit of geometric prediction losses depend entirely on the absence of pretraining, or is the effect masked by the strong backbone in the main setting?

## Statistical Analysis

*Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in the experimental section requires strengthening to support the strong claims of superiority and robustness. While the paper presents extensive benchmark results, it relies almost exclusively on point estimates (mean success rates) without accompanying measures of uncertainty.

In **Table 1 (LIBERO and LIBERO-Plus)**, success rates are reported as single percentages (e.g., 85.5% vs 84.6%). Given the sample sizes (500 rollouts per suite for LIBERO, 1 rollout per perturbed instance for LIBERO-Plus), the standard error for a 50% success rate is approximately 2.2%. The reported gain of 9.7%p in the camera-perturbation setting is substantial, but without confidence intervals or standard deviations, it is impossible to visually assess the overlap between distributions or the precision of the estimate. The authors should report mean  $\pm$  standard deviation (or 95% confidence intervals) for all aggregate metrics in Tables 1, 3, and the RoboCasa results.

Furthermore, the text in **Section 4.2** states that GAM “consistently outperforms competing baselines.” This implies statistical significance, yet no hypothesis tests are described. For binary outcomes (success/failure) across matched tasks or independent trials, appropriate tests such as McNemar’s test (for paired comparisons) or a bootstrap-based significance test should be applied. The authors should explicitly state the statistical test used and report p-values for the

key comparisons, particularly the camera-perturbation gain.

In the **Ablation Study (Table 3)**, the authors claim that removing the depth loss ( $\mathcal{L}_{\text{depth}}$ ) has “minimal impact,” citing a difference of 0.2% (89.7% vs 89.5%). Without a statistical test, this difference is indistinguishable from random noise given the variance inherent in robot learning benchmarks. The authors must provide statistical evidence (e.g., p-values from a paired test or overlapping confidence intervals) to substantiate the claim that the difference is negligible.

Finally, the **Inference Latency** comparison in **Table 4** and **Appendix Table 5** reports single latency values (e.g., 6.9ms). While latency is often deterministic, reporting the mean and standard deviation over multiple runs (e.g., 1000 iterations) would provide a more rigorous characterization of the performance gains, especially when claiming a 55x speedup.

#### Action items.

- **[science]** Report confidence intervals or standard deviations for success rates in Table 1 and Table 3. With N=500 rollouts per suite, the margin of error is ~2%, yet no uncertainty metrics are provided to assess the statistical significance of the reported 9.7%p gain.
- **[science]** Clarify the statistical test used to claim ‘consistently outperforms’ in Section 4.2. Given the binary nature of success/failure, specify if a McNemar’s test or bootstrap procedure was used to validate the differences against baselines like pi\_0.5.
- **[science]** In the ablation study (Table 3), the claim that removing  $\mathcal{L}_{\text{depth}}$  has ‘minimal impact’ lacks statistical validation. Provide p-values or effect sizes to confirm the difference between 89.7% and 89.5% is not due to random variance.

## Writing Quality

*Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_writing\_quality.md

The manuscript is generally well-written, with a clear narrative flow and precise technical terminology appropriate for the robotics and computer vision community. The abstract and introduction effectively set the stage for the proposed Geometric Action Model (GAM), and the methodological sections are logically structured. However, there are a few minor grammatical errors and typos that detract from the overall polish of the text.

Specifically, in the Appendix under “Experimental Settings and Reproducibility Details,” the sentence “The model was trained using 64 NVIDIA GH200 GPUs witch batch size of 1024” contains a clear typo (“witch” instead of “with”). Additionally, in the subsection “Generated Future Depth Maps,” the sentence “GAM predicts the and future depth maps” includes an extraneous word (“the”) that breaks the grammatical structure. In the Related Work section, the phrase “align intermediate VLA features with geometric foundation model” is missing a necessary article or pluralization, which should be corrected to “with a geometric foundation model” or “with geometric foundation models.”

While these issues are minor and do not obscure the scientific meaning, correcting them would improve the professional quality of the manuscript. The rest of the text demonstrates strong cohesion and readability, with effective use of figures and tables to support the arguments. No major structural or clarity issues were found in the main body of the paper.

**Action items.**

- **[writing]** In Appendix A (Experimental Settings), the text states ‘64 NVIDIA GH200 GPUs witch batch size of 1024’. The word ‘witch’ is a typo and should be corrected to ‘with’.
- **[writing]** In Appendix A (Generated Future Depth Maps), the sentence reads ‘GAM predicts the and future depth maps’. The word ‘the’ appears to be a typo or an extra word that disrupts the sentence flow; it should likely be removed or the sentence rephrased to ‘GAM predicts future depth maps’.
- **[writing]** In Section 3 (Related Work), the phrase ‘align intermediate VLA features with geometric foundation model’ lacks an article. It should read ‘with a geometric foundation model’ or ‘with geometric foundation models’ for grammatical correctness.